

Manish Kallu

Data Engineer | AWS · Azure · Databricks · Snowflake · Apache Spark

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PROFESSIONAL SUMMARY

Data Engineer with **5+ years** of experience delivering enterprise data platforms that support analytics, reporting, and machine learning at scale. Recognized for improving pipeline performance, reducing cloud costs, and modernizing legacy data platforms through cloud-native architectures, automation, and distributed data processing. Experienced across **AWS**, **Azure**, and modern big data ecosystems, with a strong focus on building reliable, scalable, and production-ready data solutions.

CORE TECHNICAL SKILLS

Cloud Platforms	Azure (ADF, Databricks, Synapse, ADLS Gen2, HDInsight, Cosmos DB) AWS (S3, Glue, Redshift, Kinesis, Lambda, ECS, Athena, DynamoDB) Google Cloud Platform (BigQuery, Cloud Storage)
ETL & Orchestration	Azure Data Factory, AWS Glue, Apache Airflow, dbt, Delta Lake, SSIS, Apache NiFi, REST APIs
Big Data & Streaming	Apache Spark (PySpark, Spark SQL, Spark Streaming), Apache Kafka, HDFS, Hive, HBase, Sqoop
Data Warehousing & Databases	Snowflake, Azure Synapse Analytics, Amazon Redshift, BigQuery, Teradata, Oracle, PostgreSQL
Programming	Python, SQL, Scala, Java, T-SQL, PL/SQL, Bash
DevOps & Containers	Git, GitHub, Jenkins, Terraform, CloudFormation, Docker, Kubernetes, Amazon ECS
BI, Modeling & Governance	Power BI, Tableau, Collibra, Data Modeling (Star Schema, Snowflake Schema, SCD Type I/II), Erwin Data Modeler

PROFESSIONAL EXPERIENCE

Data Engineer II

HCLTech, India

Oct 2023 – Jun 2025

- Architected and maintained **60+ enterprise-grade batch and streaming data pipelines**, processing **150–300 million records per pipeline execution** across AWS and Azure cloud platforms for analytics, regulatory reporting, and business intelligence.
- Designed and implemented end-to-end Lakehouse architecture using Azure Data Factory, Azure Databricks, Delta Lake, and ADLS Gen2, enabling ACID transactions, schema evolution, and time-travel capabilities.
- Modernized legacy Oracle and SQL Server ETL workloads by migrating to Azure Synapse Analytics and Databricks PySpark pipelines, reducing execution time by over **40%** while improving scalability.
- Developed modular **dbt** models on Snowflake and Azure Synapse, implementing automated testing, documentation, lineage, and reusable SQL transformations.
- Optimized Spark, Databricks, and Delta Lake workloads using partitioning, caching, Adaptive Query Execution (AQE), broadcast joins, and file compaction, reducing processing time by nearly **50%**.
- Designed scalable feature engineering pipelines using PySpark and SQL to support machine learning workflows and analytical model development.

- Integrated machine learning prediction outputs into batch and near real-time ETL pipelines, enabling downstream reporting and operational analytics.
- Implemented Medallion Architecture (Bronze, Silver, Gold) using Delta Lake to improve data quality, governance, and downstream consumption across enterprise data products.
- Developed reusable Python and PySpark frameworks for metadata-driven ingestion, transformation, validation, and standardized ETL development.
- Built automated data quality frameworks incorporating schema validation, null handling, duplicate detection, reconciliation, and business rule enforcement.
- Designed enterprise ELT pipelines integrating Snowflake with AWS and Azure data platforms to support scalable analytics and self-service reporting.
- Implemented observability and monitoring using Amazon CloudWatch, Databricks monitoring, and centralized logging, reducing incident detection and troubleshooting time by over **60%**.
- Optimized cloud infrastructure utilization across AWS and Azure services through workload tuning, auto-scaling, and resource optimization, reducing operational costs by approximately **25%**.
- Established CI/CD pipelines using Azure DevOps, Git, Jenkins, and Infrastructure as Code practices to automate deployment, testing, and release management of data engineering workflows.
- Built reusable ingestion frameworks using Azure Data Factory, AWS Glue, and Databricks, reducing onboarding time for new data sources by over **40%**.
- Implemented enterprise data governance, metadata management, lineage tracking, and data quality controls using dbt and Collibra to improve compliance and audit readiness.
- Developed AWS Glue ETL jobs using PySpark to ingest structured and semi-structured datasets from Amazon S3 into Amazon Redshift, leveraging Glue Crawlers and Data Catalog for automated metadata management.
- Built high-throughput real-time streaming pipelines using Amazon Kinesis Data Streams, Kinesis Firehose, Apache Kafka, and Spark Structured Streaming for event-driven analytics.
- Deployed and managed containerized ETL workloads using Docker on Amazon ECS and Kubernetes-based environments, implementing auto-scaling, task orchestration, and resilient pipeline execution.
- Developed advanced data transformation and enrichment workflows using Python (Pandas, PySpark) and Spark SQL to process large-scale structured and semi-structured datasets.
- Developed AWS Lambda functions for event-driven orchestration, metadata processing, and automated ETL workflows triggered by Amazon S3 events and scheduled CloudWatch rules.
- Configured enterprise monitoring, IAM least-privilege access, encryption, auditing, and security best practices across AWS and Azure data engineering environments.
- Collaborated with solution architects, data scientists, platform engineers, and business stakeholders to define data contracts, governance standards, SLAs, and scalable enterprise data platform solutions.

Stack: Databricks · Apache Spark (PySpark, Spark SQL) · Delta Lake · dbt · Snowflake · Azure Data Factory (ADF) · Azure Synapse Analytics · AWS Glue · Amazon S3 · Amazon Redshift · Amazon Kinesis · AWS Lambda · Docker · Amazon ECS · Kubernetes · Python · Apache Airflow

Data Engineer
HCLTech, India

Jan 2021 – Sep 2023

- Designed, developed, and maintained **35+ production ETL/ELT pipelines** using Azure Data Factory and Azure Synapse Analytics, processing **25–40 million records per pipeline execution** for regulatory reporting and enterprise analytics.
- Developed Azure Functions to automate schema generation and event-driven data transformations from Azure Blob Storage and Azure Cosmos DB into PostgreSQL using serverless architectures.
- Built and maintained SSIS packages integrating Oracle, SQL Server, Azure SQL Database, Excel, REST APIs, and flat files into Azure SQL Data Warehouse and Azure Synapse Analytics.
- Developed near real-time streaming ingestion pipelines using Apache Kafka, Kafka REST Proxy, Spark Structured Streaming, and HDFS to process high-volume transactional and application event data.

- Optimized Spark SQL, Hive, and Synapse workloads through partitioning, bucketing, caching, indexing, and query tuning, reducing execution times by nearly **30%**.
- Automated operational workflows and metadata-driven ETL processes using Python, significantly reducing manual effort and improving deployment consistency.
- Implemented centralized logging, exception handling, retry mechanisms, and automated alerting across Python and ADF pipelines, improving operational reliability.
- Designed and implemented cloud-native data ingestion pipelines using Google Cloud Storage and BigQuery for batch and analytical workloads.
- Developed scalable ingestion frameworks to load structured, semi-structured, and JSON datasets into BigQuery using Python, Cloud Storage, and scheduled workflows.
- Performed large-scale data validation, reconciliation, profiling, and ad-hoc analytics using BigQuery SQL to support business intelligence and data quality initiatives.
- Integrated Azure Data Factory with AWS S3 and Google Cloud Storage, enabling secure cross-cloud data movement and hybrid analytics solutions.
- Designed reusable metadata-driven ETL frameworks in Azure Data Factory and SSIS, accelerating development and improving pipeline standardization.
- Implemented end-to-end data validation, reconciliation, auditing, and quality checks to ensure regulatory compliance and reporting accuracy.
- Automated ingestion, transformation, scheduling, and monitoring workflows using Azure Data Factory triggers and Python automation scripts.
- Developed optimized HiveQL scripts for aggregation, joins, partition pruning, bucketing, and warehouse optimization across large-scale analytical datasets.
- Built CI/CD pipelines using Git, Azure DevOps, Jenkins, and Python scripting for automated deployment of ETL workflows and infrastructure changes.
- Developed Power BI dashboards and semantic models using Azure SQL, Synapse, and T-SQL stored procedures to deliver interactive business and operational reporting.
- Collaborated with business analysts, architects, QA teams, and product stakeholders to gather requirements, define SLAs, and deliver reliable enterprise data solutions.
- Tuned Azure SQL Database, Synapse Analytics, and PostgreSQL workloads by optimizing indexing strategies, execution plans, and resource utilization for high-performance analytics.
- Managed pipeline scheduling, dependency management, monitoring, and failure recovery using Azure Data Factory, improving overall pipeline availability and operational efficiency.
- Built scalable Spark applications using PySpark, Spark SQL, and DataFrames to transform large datasets exceeding **500 million records per day** across distributed environments.
- Implemented Git-based version control, code reviews, branching strategies, and documentation standards to improve collaboration and release quality.
- Worked with dimensional data models, star schemas, slowly changing dimensions (SCD Type I & II), and fact/dimension modeling to support enterprise reporting and analytics.

Stack: Azure Data Factory (ADF) · Azure Synapse Analytics · Azure Data Lake Storage Gen2 (ADLS Gen2) · Azure SQL Database · Apache Spark (PySpark, Spark SQL) · Apache Kafka · Hive · BigQuery · Google Cloud Storage · Python · SQL · Power BI · Git · Jenkins · Azure DevOps

Intern - Data Engineer
HCLTech, India

Sep 2019 – Dec 2020

- Assisted in building AWS-native data pipelines integrating Amazon S3, AWS Glue (PySpark), Amazon Redshift, and Snowflake for batch data ingestion and analytical reporting.
- Developed Snowflake ingestion pipelines using SnowSQL and Snowpipe to load structured and semi-structured JSON data from Amazon S3.
- Implemented streaming data pipelines using Apache Kafka and Spark Structured Streaming to process simulated application and log event data.

- Processed large-scale datasets using Apache Spark and AWS Glue, performing data cleansing, transformations, and aggregations.
- Developed interactive Power BI dashboards to visualize business KPIs, trends, and operational metrics from curated datasets.
- Created dimensional data models and DAX measures in Power BI to enable dynamic reporting and drill-through analysis.
- Connected Power BI to Snowflake, Amazon Redshift, and SQL-based data sources to deliver interactive business insights.
- Worked with team members to understand reporting requirements and translate business needs into analytical dashboards.
- Improved ETL workflow performance by optimizing Spark transformations, partitioning strategies, and minimizing redundant operations.
- Developed reusable PySpark modules for data ingestion, transformation, and validation to improve code maintainability.
- Applied partitioning and file optimization techniques in Amazon S3 and Spark to improve query performance and storage efficiency.
- Enhanced Kafka consumer applications by implementing efficient message processing and error-handling mechanisms.
- Assisted in modernizing traditional ETL workflows using cloud-native AWS services and Snowflake-based data pipelines.
- Built scalable data pipelines using Databricks notebooks and Spark for distributed batch processing and transformation.
- Developed Spark workloads in Databricks using PySpark DataFrames and Spark SQL for batch analytics.
- Integrated Amazon S3 with Snowflake and Databricks to support end-to-end ELT workflows.
- Optimized Snowflake SQL queries using clustering, pruning techniques, and warehouse tuning to improve analytical performance.
- Automated end-to-end data workflows using Apache Airflow DAGs for orchestration and scheduling.
- Developed Spark applications using PySpark, DataFrames, and Spark SQL for both batch and near real-time processing scenarios.
- Designed and maintained Apache Airflow DAGs to orchestrate data movement between Amazon S3, AWS Glue, Databricks, and Snowflake.
- Developed dbt models to transform curated Snowflake datasets and implement modular ELT workflows with testing and documentation.
- Provisioned AWS resources using Terraform and automated deployment workflows through Git-based version control practices.
- Performed data quality validation, debugging, and monitoring across Spark, Snowflake, and Airflow pipelines to ensure reliable data delivery.

Stack: AWS (S3, EC2, Glue, Redshift, Athena, EMR) · Snowflake · Apache Spark · Kafka · NiFi · Airflow · Hive · HBase · dbt · Terraform · Jenkins · Python · Scala · Tableau

EDUCATION

Master of Science in Data Science
 Willamette University
 Salem, OR, United States

Expected : Aug 2026
GPA : 3.85 / 4.0

Coursework: Data Engineering, Machine Learning, Statistical Modeling, Big Data Analytics, Data Visualization, Cloud Computing, Database Management Systems, Deep Learning, Data Mining, Python for Data Science, Applied Artificial Intelligence, Predictive Analytics, Natural Language Processing, and Distributed Computing.